

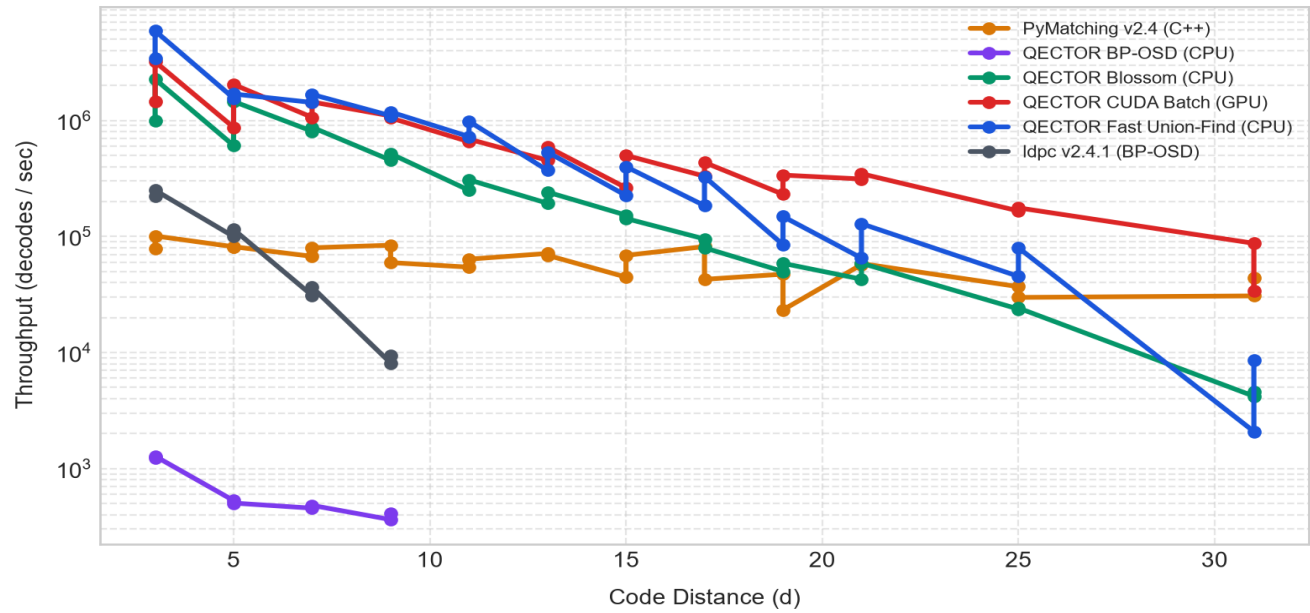
QECTOR v0.7.0 Complete Benchmark Report

Full Untruncated Data: Distances $d=3..31$ | Shot Volumes $S=1k..100k$ | QECTOR CPU/GPU vs PyMatching & Idpc

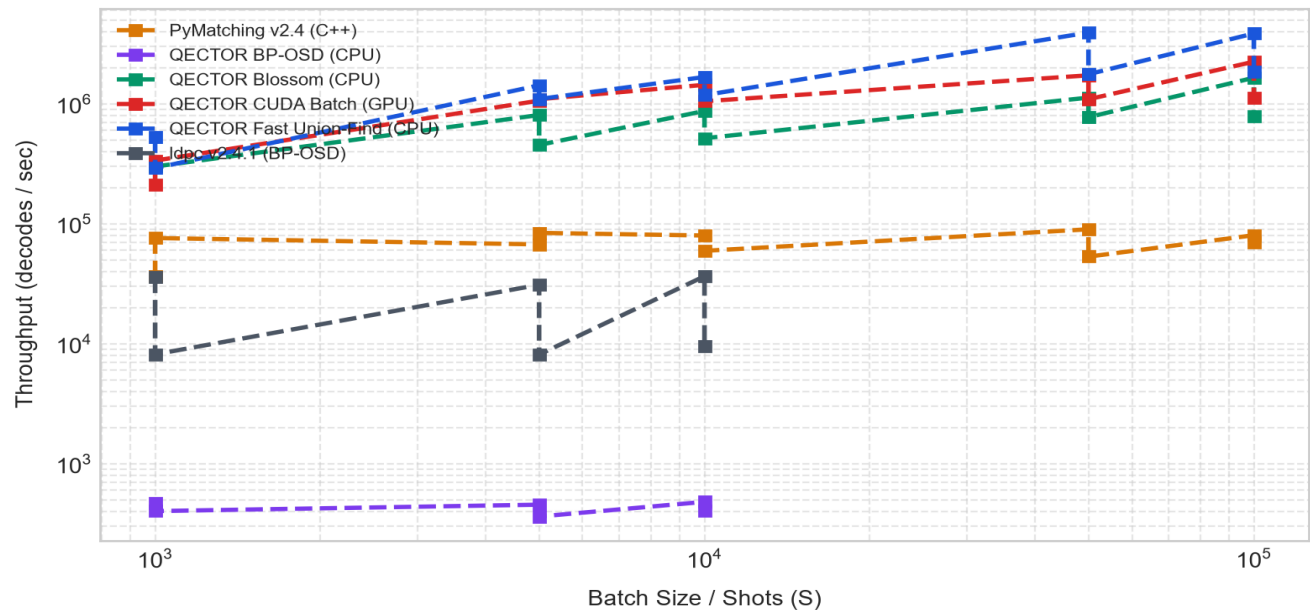
Generated UTC: 2026-07-31T20:49:18 | Git Commit: 3d60a96e44 | QECTOR: v0.7.0
PyMatching: v2.4.0 | Idpc: v2.4.1 | Stim: v1.16.0 | Platform: Windows-10-10.0.26100-SP0

Visual Performance Scaling (Throughput & Batch Volume)

Decoding Throughput vs Code Distance ($d=3..31$)



Decoding Throughput vs Batch Size S (1k..100k)



Complete Empirical Benchmark Table (264 Configurations)

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR Fast Union-Find (CPU)	QECTOR CPU	3	1,000	305,782.3 dec/s	3.27 μs	3.84x
QECTOR Blossom (CPU)	QECTOR CPU	3	1,000	423,710.9 dec/s	2.36 μs	5.32x
QECTOR BP-OSD (CPU)	QECTOR CPU	3	1,000	102.9 dec/s	9713.60 μs	0.0x
QECTOR CUDA Batch (GPU)	QECTOR GPU	3	1,000	384,083.6 dec/s	2.60 μs	4.82x
PyMatching v2.4 (C++)	PyMatching	3	1,000	79,652.7 dec/s	12.55 μs	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	3	1,000	187,740.5 dec/s	5.33 μs	2.36x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	3	5,000	3,429,120.0 dec/s	0.29 μs	43.85x
QECTOR Blossom (CPU)	QECTOR CPU	3	5,000	996,850.0 dec/s	1.00 μs	12.75x
QECTOR BP-OSD (CPU)	QECTOR CPU	3	5,000	1,222.7 dec/s	817.86 μs	0.02x
QECTOR CUDA Batch (GPU)	QECTOR GPU	3	5,000	1,446,717.4 dec/s	0.69 μs	18.5x
PyMatching v2.4 (C++)	PyMatching	3	5,000	78,204.4 dec/s	12.79 μs	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	3	5,000	223,388.8 dec/s	4.48 μs	2.86x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	3	10,000	5,903,536.3 dec/s	0.17 μs	58.52x
QECTOR Blossom (CPU)	QECTOR CPU	3	10,000	2,264,595.3 dec/s	0.44 μs	22.45x
QECTOR BP-OSD (CPU)	QECTOR CPU	3	10,000	1,277.4 dec/s	782.84 μs	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	3	10,000	3,199,897.7 dec/s	0.31 μs	31.72x
PyMatching v2.4 (C++)	PyMatching	3	10,000	100,877.6 dec/s	9.91 μs	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	3	10,000	250,062.5 dec/s	4.00 μs	2.48x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	3	50,000	7,998,464.3 dec/s	0.13 μs	126.15x
QECTOR Blossom (CPU)	QECTOR CPU	3	50,000	4,109,848.0 dec/s	0.24 μs	64.82x
QECTOR CUDA Batch (GPU)	QECTOR GPU	3	50,000	7,878,606.4 dec/s	0.13 μs	124.26x
PyMatching v2.4 (C++)	PyMatching	3	50,000	63,405.5 dec/s	15.77 μs	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	3	100,000	8,043,111.1 dec/s	0.12 μs	85.84x
QECTOR Blossom (CPU)	QECTOR CPU	3	100,000	5,784,693.7 dec/s	0.17 μs	61.73x
QECTOR CUDA Batch (GPU)	QECTOR GPU	3	100,000	10,727,426.8 dec/s	0.09 μs	114.48x
PyMatching v2.4 (C++)	PyMatching	3	100,000	93,703.1 dec/s	10.67 μs	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	5	1,000	968,241.7 dec/s	1.03 μs	20.82x
QECTOR Blossom (CPU)	QECTOR CPU	5	1,000	186,954.3 dec/s	5.35 μs	4.02x
QECTOR BP-OSD (CPU)	QECTOR CPU	5	1,000	530.1 dec/s	1886.54 μs	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	5	1,000	268,175.6 dec/s	3.73 μs	5.77x
PyMatching v2.4 (C++)	PyMatching	5	1,000	46,501.9 dec/s	21.50 μs	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	5	1,000	126,614.3 dec/s	7.90 μs	2.72x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	5	5,000	1,555,645.4 dec/s	0.64 μs	19.09x
QECTOR Blossom (CPU)	QECTOR CPU	5	5,000	607,208.8 dec/s	1.65 μs	7.45x
QECTOR BP-OSD (CPU)	QECTOR CPU	5	5,000	529.1 dec/s	1889.92 μs	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	5	5,000	870,367.5 dec/s	1.15 μs	10.68x
PyMatching v2.4 (C++)	PyMatching	5	5,000	81,469.7 dec/s	12.27 μs	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	5	5,000	99,700.9 dec/s	10.03 μs	1.22x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	5	10,000	1,688,020.1 dec/s	0.59 μs	20.72x
QECTOR Blossom (CPU)	QECTOR CPU	5	10,000	1,455,265.2 dec/s	0.69 μs	17.86x
QECTOR BP-OSD (CPU)	QECTOR CPU	5	10,000	503.4 dec/s	1986.39 μs	0.01x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR CUDA Batch (GPU)	QECTOR GPU	5	10,000	2,038,195.8 dec/s	0.49 μ s	25.01x
PyMatching v2.4 (C++)	PyMatching	5	10,000	81,483.0 dec/s	12.27 μ s	1.0x
ldpc v2.4.1 (BP-OSD)	ldpc	5	10,000	114,817.2 dec/s	8.71 μ s	1.41x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	5	50,000	5,261,219.5 dec/s	0.19 μ s	69.74x
QECTOR Blossom (CPU)	QECTOR CPU	5	50,000	2,005,615.7 dec/s	0.50 μ s	26.59x
QECTOR CUDA Batch (GPU)	QECTOR GPU	5	50,000	3,199,467.6 dec/s	0.31 μ s	42.41x
PyMatching v2.4 (C++)	PyMatching	5	50,000	75,437.5 dec/s	13.26 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	5	100,000	6,155,626.5 dec/s	0.16 μ s	66.46x
QECTOR Blossom (CPU)	QECTOR CPU	5	100,000	2,989,161.3 dec/s	0.33 μ s	32.27x
QECTOR CUDA Batch (GPU)	QECTOR GPU	5	100,000	3,202,756.9 dec/s	0.31 μ s	34.58x
PyMatching v2.4 (C++)	PyMatching	5	100,000	92,622.6 dec/s	10.80 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	7	1,000	526,509.8 dec/s	1.90 μ s	14.47x
QECTOR Blossom (CPU)	QECTOR CPU	7	1,000	313,755.0 dec/s	3.19 μ s	8.62x
QECTOR BP-OSD (CPU)	QECTOR CPU	7	1,000	467.0 dec/s	2141.52 μ s	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	7	1,000	210,009.0 dec/s	4.76 μ s	5.77x
PyMatching v2.4 (C++)	PyMatching	7	1,000	36,382.2 dec/s	27.49 μ s	1.0x
ldpc v2.4.1 (BP-OSD)	ldpc	7	1,000	35,807.6 dec/s	27.93 μ s	0.98x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	7	5,000	1,426,818.5 dec/s	0.70 μ s	21.2x
QECTOR Blossom (CPU)	QECTOR CPU	7	5,000	806,308.5 dec/s	1.24 μ s	11.98x
QECTOR BP-OSD (CPU)	QECTOR CPU	7	5,000	455.1 dec/s	2197.16 μ s	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	7	5,000	1,062,789.6 dec/s	0.94 μ s	15.79x
PyMatching v2.4 (C++)	PyMatching	7	5,000	67,317.4 dec/s	14.85 μ s	1.0x
ldpc v2.4.1 (BP-OSD)	ldpc	7	5,000	31,000.5 dec/s	32.26 μ s	0.46x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	7	10,000	1,673,444.1 dec/s	0.60 μ s	20.97x
QECTOR Blossom (CPU)	QECTOR CPU	7	10,000	871,938.4 dec/s	1.15 μ s	10.92x
QECTOR BP-OSD (CPU)	QECTOR CPU	7	10,000	480.9 dec/s	2079.49 μ s	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	7	10,000	1,440,278.8 dec/s	0.69 μ s	18.05x
PyMatching v2.4 (C++)	PyMatching	7	10,000	79,814.8 dec/s	12.53 μ s	1.0x
ldpc v2.4.1 (BP-OSD)	ldpc	7	10,000	36,513.0 dec/s	27.39 μ s	0.46x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	7	50,000	3,927,483.0 dec/s	0.25 μ s	43.67x
QECTOR Blossom (CPU)	QECTOR CPU	7	50,000	1,126,671.7 dec/s	0.89 μ s	12.53x
QECTOR CUDA Batch (GPU)	QECTOR GPU	7	50,000	1,727,796.1 dec/s	0.58 μ s	19.21x
PyMatching v2.4 (C++)	PyMatching	7	50,000	89,932.1 dec/s	11.12 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	7	100,000	3,859,111.6 dec/s	0.26 μ s	48.08x
QECTOR Blossom (CPU)	QECTOR CPU	7	100,000	1,649,764.8 dec/s	0.61 μ s	20.56x
QECTOR CUDA Batch (GPU)	QECTOR GPU	7	100,000	2,245,072.1 dec/s	0.45 μ s	27.97x
PyMatching v2.4 (C++)	PyMatching	7	100,000	80,260.0 dec/s	12.46 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	9	1,000	290,985.3 dec/s	3.44 μ s	3.82x
QECTOR Blossom (CPU)	QECTOR CPU	9	1,000	300,246.2 dec/s	3.33 μ s	3.95x
QECTOR BP-OSD (CPU)	QECTOR CPU	9	1,000	402.4 dec/s	2484.96 μ s	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	9	1,000	336,394.5 dec/s	2.97 μ s	4.42x
PyMatching v2.4 (C++)	PyMatching	9	1,000	76,077.4 dec/s	13.14 μ s	1.0x
ldpc v2.4.1 (BP-OSD)	ldpc	9	1,000	8,156.5 dec/s	122.60 μ s	0.11x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR Fast Union-Find (CPU)	QECTOR CPU	9	5,000	1,096,683.7 dec/s	0.91 μ s	13.06x
QECTOR Blossom (CPU)	QECTOR CPU	9	5,000	453,831.7 dec/s	2.20 μ s	5.41x
QECTOR BP-OSD (CPU)	QECTOR CPU	9	5,000	362.6 dec/s	2757.73 μ s	0.0x
QECTOR CUDA Batch (GPU)	QECTOR GPU	9	5,000	1,093,302.4 dec/s	0.91 μ s	13.02x
PyMatching v2.4 (C++)	PyMatching	9	5,000	83,963.1 dec/s	11.91 μ s	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	9	5,000	8,123.0 dec/s	123.11 μ s	0.1x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	9	10,000	1,181,348.9 dec/s	0.85 μ s	19.86x
QECTOR Blossom (CPU)	QECTOR CPU	9	10,000	516,406.2 dec/s	1.94 μ s	8.68x
QECTOR BP-OSD (CPU)	QECTOR CPU	9	10,000	407.5 dec/s	2453.86 μ s	0.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	9	10,000	1,050,376.0 dec/s	0.95 μ s	17.66x
PyMatching v2.4 (C++)	PyMatching	9	10,000	59,488.4 dec/s	16.81 μ s	1.0x
Idpc v2.4.1 (BP-OSD)	Idpc	9	10,000	9,468.0 dec/s	105.62 μ s	0.16x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	9	50,000	1,765,112.9 dec/s	0.57 μ s	33.16x
QECTOR Blossom (CPU)	QECTOR CPU	9	50,000	773,804.5 dec/s	1.29 μ s	14.54x
QECTOR CUDA Batch (GPU)	QECTOR GPU	9	50,000	1,091,526.7 dec/s	0.92 μ s	20.51x
PyMatching v2.4 (C++)	PyMatching	9	50,000	53,231.1 dec/s	18.79 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	9	100,000	1,862,610.2 dec/s	0.54 μ s	26.51x
QECTOR Blossom (CPU)	QECTOR CPU	9	100,000	792,684.5 dec/s	1.26 μ s	11.28x
QECTOR CUDA Batch (GPU)	QECTOR GPU	9	100,000	1,126,490.2 dec/s	0.89 μ s	16.03x
PyMatching v2.4 (C++)	PyMatching	9	100,000	70,271.6 dec/s	14.23 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	11	1,000	158,353.1 dec/s	6.32 μ s	5.54x
QECTOR Blossom (CPU)	QECTOR CPU	11	1,000	232,628.5 dec/s	4.30 μ s	8.14x
QECTOR CUDA Batch (GPU)	QECTOR GPU	11	1,000	119,838.0 dec/s	8.34 μ s	4.19x
PyMatching v2.4 (C++)	PyMatching	11	1,000	28,594.3 dec/s	34.97 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	11	5,000	724,364.7 dec/s	1.38 μ s	13.33x
QECTOR Blossom (CPU)	QECTOR CPU	11	5,000	249,591.9 dec/s	4.01 μ s	4.59x
QECTOR CUDA Batch (GPU)	QECTOR GPU	11	5,000	654,501.7 dec/s	1.53 μ s	12.04x
PyMatching v2.4 (C++)	PyMatching	11	5,000	54,341.9 dec/s	18.40 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	11	10,000	980,315.3 dec/s	1.02 μ s	15.39x
QECTOR Blossom (CPU)	QECTOR CPU	11	10,000	307,169.6 dec/s	3.26 μ s	4.82x
QECTOR CUDA Batch (GPU)	QECTOR GPU	11	10,000	687,989.8 dec/s	1.45 μ s	10.8x
PyMatching v2.4 (C++)	PyMatching	11	10,000	63,678.0 dec/s	15.70 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	11	50,000	1,119,667.9 dec/s	0.89 μ s	27.04x
QECTOR Blossom (CPU)	QECTOR CPU	11	50,000	466,012.3 dec/s	2.15 μ s	11.26x
QECTOR CUDA Batch (GPU)	QECTOR GPU	11	50,000	458,865.9 dec/s	2.18 μ s	11.08x
PyMatching v2.4 (C++)	PyMatching	11	50,000	41,404.4 dec/s	24.15 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	11	100,000	1,336,482.1 dec/s	0.75 μ s	27.15x
QECTOR Blossom (CPU)	QECTOR CPU	11	100,000	513,235.8 dec/s	1.95 μ s	10.42x
QECTOR CUDA Batch (GPU)	QECTOR GPU	11	100,000	562,781.7 dec/s	1.78 μ s	11.43x
PyMatching v2.4 (C++)	PyMatching	11	100,000	49,232.0 dec/s	20.31 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	13	1,000	83,015.8 dec/s	12.05 μ s	2.75x
QECTOR Blossom (CPU)	QECTOR CPU	13	1,000	93,936.4 dec/s	10.65 μ s	3.11x
QECTOR CUDA Batch (GPU)	QECTOR GPU	13	1,000	94,512.6 dec/s	10.58 μ s	3.13x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
PyMatching v2.4 (C++)	PyMatching	13	1,000	30,202.4 dec/s	33.11 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	13	5,000	370,895.1 dec/s	2.70 μ s	5.2x
QECTOR Blossom (CPU)	QECTOR CPU	13	5,000	192,991.3 dec/s	5.18 μ s	2.71x
QECTOR CUDA Batch (GPU)	QECTOR GPU	13	5,000	452,878.0 dec/s	2.21 μ s	6.35x
PyMatching v2.4 (C++)	PyMatching	13	5,000	71,316.5 dec/s	14.02 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	13	10,000	532,594.8 dec/s	1.88 μ s	7.75x
QECTOR Blossom (CPU)	QECTOR CPU	13	10,000	240,956.9 dec/s	4.15 μ s	3.5x
QECTOR CUDA Batch (GPU)	QECTOR GPU	13	10,000	590,113.2 dec/s	1.69 μ s	8.58x
PyMatching v2.4 (C++)	PyMatching	13	10,000	68,747.4 dec/s	14.55 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	13	50,000	850,976.9 dec/s	1.18 μ s	13.38x
QECTOR Blossom (CPU)	QECTOR CPU	13	50,000	253,510.5 dec/s	3.94 μ s	3.99x
QECTOR CUDA Batch (GPU)	QECTOR GPU	13	50,000	337,733.1 dec/s	2.96 μ s	5.31x
PyMatching v2.4 (C++)	PyMatching	13	50,000	63,580.9 dec/s	15.73 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	13	100,000	882,815.1 dec/s	1.13 μ s	8.66x
QECTOR Blossom (CPU)	QECTOR CPU	13	100,000	239,409.7 dec/s	4.18 μ s	2.35x
QECTOR CUDA Batch (GPU)	QECTOR GPU	13	100,000	446,470.0 dec/s	2.24 μ s	4.38x
PyMatching v2.4 (C++)	PyMatching	13	100,000	101,957.6 dec/s	9.81 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	15	1,000	90,916.5 dec/s	11.00 μ s	3.63x
QECTOR Blossom (CPU)	QECTOR CPU	15	1,000	55,870.0 dec/s	17.90 μ s	2.23x
QECTOR CUDA Batch (GPU)	QECTOR GPU	15	1,000	146,331.5 dec/s	6.83 μ s	5.84x
PyMatching v2.4 (C++)	PyMatching	15	1,000	25,050.1 dec/s	39.92 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	15	5,000	225,266.8 dec/s	4.44 μ s	5.05x
QECTOR Blossom (CPU)	QECTOR CPU	15	5,000	152,546.9 dec/s	6.56 μ s	3.42x
QECTOR CUDA Batch (GPU)	QECTOR GPU	15	5,000	262,472.7 dec/s	3.81 μ s	5.88x
PyMatching v2.4 (C++)	PyMatching	15	5,000	44,607.0 dec/s	22.42 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	15	10,000	398,942.0 dec/s	2.51 μ s	5.81x
QECTOR Blossom (CPU)	QECTOR CPU	15	10,000	142,717.7 dec/s	7.01 μ s	2.08x
QECTOR CUDA Batch (GPU)	QECTOR GPU	15	10,000	499,390.7 dec/s	2.00 μ s	7.27x
PyMatching v2.4 (C++)	PyMatching	15	10,000	68,719.1 dec/s	14.55 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	15	50,000	622,015.1 dec/s	1.61 μ s	5.76x
QECTOR Blossom (CPU)	QECTOR CPU	15	50,000	162,173.8 dec/s	6.17 μ s	1.5x
QECTOR CUDA Batch (GPU)	QECTOR GPU	15	50,000	350,567.7 dec/s	2.85 μ s	3.25x
PyMatching v2.4 (C++)	PyMatching	15	50,000	107,898.1 dec/s	9.27 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	15	100,000	612,141.6 dec/s	1.63 μ s	8.98x
QECTOR Blossom (CPU)	QECTOR CPU	15	100,000	161,444.1 dec/s	6.19 μ s	2.37x
QECTOR CUDA Batch (GPU)	QECTOR GPU	15	100,000	309,200.0 dec/s	3.23 μ s	4.53x
PyMatching v2.4 (C++)	PyMatching	15	100,000	68,194.2 dec/s	14.66 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	17	1,000	49,375.6 dec/s	20.25 μ s	2.9x
QECTOR Blossom (CPU)	QECTOR CPU	17	1,000	62,016.7 dec/s	16.12 μ s	3.64x
QECTOR CUDA Batch (GPU)	QECTOR GPU	17	1,000	118,842.5 dec/s	8.41 μ s	6.98x
PyMatching v2.4 (C++)	PyMatching	17	1,000	17,014.3 dec/s	58.77 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	17	5,000	184,103.1 dec/s	5.43 μ s	2.25x
QECTOR Blossom (CPU)	QECTOR CPU	17	5,000	95,042.8 dec/s	10.52 μ s	1.16x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR CUDA Batch (GPU)	QECTOR GPU	17	5,000	330,517.3 dec/s	3.03 μ s	4.04x
PyMatching v2.4 (C++)	PyMatching	17	5,000	81,739.4 dec/s	12.23 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	17	10,000	327,010.0 dec/s	3.06 μ s	7.66x
QECTOR Blossom (CPU)	QECTOR CPU	17	10,000	79,711.6 dec/s	12.55 μ s	1.87x
QECTOR CUDA Batch (GPU)	QECTOR GPU	17	10,000	433,954.3 dec/s	2.30 μ s	10.17x
PyMatching v2.4 (C++)	PyMatching	17	10,000	42,684.0 dec/s	23.43 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	17	50,000	457,543.6 dec/s	2.19 μ s	16.84x
QECTOR Blossom (CPU)	QECTOR CPU	17	50,000	84,536.8 dec/s	11.83 μ s	3.11x
QECTOR CUDA Batch (GPU)	QECTOR GPU	17	50,000	205,502.3 dec/s	4.87 μ s	7.56x
PyMatching v2.4 (C++)	PyMatching	17	50,000	27,168.0 dec/s	36.81 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	17	100,000	368,013.9 dec/s	2.72 μ s	6.65x
QECTOR Blossom (CPU)	QECTOR CPU	17	100,000	67,906.7 dec/s	14.73 μ s	1.23x
QECTOR CUDA Batch (GPU)	QECTOR GPU	17	100,000	284,579.6 dec/s	3.51 μ s	5.14x
PyMatching v2.4 (C++)	PyMatching	17	100,000	55,328.1 dec/s	18.07 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	19	1,000	43,884.5 dec/s	22.79 μ s	5.38x
QECTOR Blossom (CPU)	QECTOR CPU	19	1,000	43,272.1 dec/s	23.11 μ s	5.31x
QECTOR CUDA Batch (GPU)	QECTOR GPU	19	1,000	74,540.6 dec/s	13.42 μ s	9.15x
PyMatching v2.4 (C++)	PyMatching	19	1,000	8,150.6 dec/s	122.69 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	19	5,000	84,951.2 dec/s	11.77 μ s	1.8x
QECTOR Blossom (CPU)	QECTOR CPU	19	5,000	49,924.8 dec/s	20.03 μ s	1.06x
QECTOR CUDA Batch (GPU)	QECTOR GPU	19	5,000	230,892.5 dec/s	4.33 μ s	4.89x
PyMatching v2.4 (C++)	PyMatching	19	5,000	47,259.0 dec/s	21.16 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	19	10,000	148,528.4 dec/s	6.73 μ s	6.38x
QECTOR Blossom (CPU)	QECTOR CPU	19	10,000	58,605.0 dec/s	17.06 μ s	2.52x
QECTOR CUDA Batch (GPU)	QECTOR GPU	19	10,000	337,474.1 dec/s	2.96 μ s	14.51x
PyMatching v2.4 (C++)	PyMatching	19	10,000	23,263.9 dec/s	42.98 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	19	50,000	298,036.1 dec/s	3.36 μ s	6.19x
QECTOR Blossom (CPU)	QECTOR CPU	19	50,000	66,733.4 dec/s	14.98 μ s	1.39x
QECTOR CUDA Batch (GPU)	QECTOR GPU	19	50,000	175,751.9 dec/s	5.69 μ s	3.65x
PyMatching v2.4 (C++)	PyMatching	19	50,000	48,181.2 dec/s	20.75 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	19	100,000	357,352.5 dec/s	2.80 μ s	4.89x
QECTOR Blossom (CPU)	QECTOR CPU	19	100,000	67,008.6 dec/s	14.92 μ s	0.92x
QECTOR CUDA Batch (GPU)	QECTOR GPU	19	100,000	163,862.1 dec/s	6.10 μ s	2.24x
PyMatching v2.4 (C++)	PyMatching	19	100,000	73,019.4 dec/s	13.69 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	21	1,000	37,574.2 dec/s	26.61 μ s	4.59x
QECTOR Blossom (CPU)	QECTOR CPU	21	1,000	29,881.4 dec/s	33.47 μ s	3.65x
QECTOR CUDA Batch (GPU)	QECTOR GPU	21	1,000	45,595.3 dec/s	21.93 μ s	5.56x
PyMatching v2.4 (C++)	PyMatching	21	1,000	8,194.4 dec/s	122.04 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	21	5,000	65,058.0 dec/s	15.37 μ s	1.14x
QECTOR Blossom (CPU)	QECTOR CPU	21	5,000	42,781.8 dec/s	23.37 μ s	0.75x
QECTOR CUDA Batch (GPU)	QECTOR GPU	21	5,000	312,431.7 dec/s	3.20 μ s	5.47x
PyMatching v2.4 (C++)	PyMatching	21	5,000	57,159.2 dec/s	17.49 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	21	10,000	128,637.2 dec/s	7.77 μ s	2.21x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR Blossom (CPU)	QECTOR CPU	21	10,000	58,702.9 dec/s	17.03 μ s	1.01x
QECTOR CUDA Batch (GPU)	QECTOR GPU	21	10,000	346,804.4 dec/s	2.88 μ s	5.95x
PyMatching v2.4 (C++)	PyMatching	21	10,000	58,326.1 dec/s	17.14 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	21	50,000	354,371.7 dec/s	2.82 μ s	6.0x
QECTOR Blossom (CPU)	QECTOR CPU	21	50,000	60,379.6 dec/s	16.56 μ s	1.02x
QECTOR CUDA Batch (GPU)	QECTOR GPU	21	50,000	160,213.1 dec/s	6.24 μ s	2.71x
PyMatching v2.4 (C++)	PyMatching	21	50,000	59,049.3 dec/s	16.94 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	21	100,000	318,659.5 dec/s	3.14 μ s	5.58x
QECTOR Blossom (CPU)	QECTOR CPU	21	100,000	57,306.2 dec/s	17.45 μ s	1.0x
QECTOR CUDA Batch (GPU)	QECTOR GPU	21	100,000	42,521.6 dec/s	23.52 μ s	0.74x
PyMatching v2.4 (C++)	PyMatching	21	100,000	57,126.5 dec/s	17.50 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	25	1,000	30,981.9 dec/s	32.28 μ s	4.6x
QECTOR Blossom (CPU)	QECTOR CPU	25	1,000	19,570.0 dec/s	51.10 μ s	2.91x
QECTOR CUDA Batch (GPU)	QECTOR GPU	25	1,000	31,202.8 dec/s	32.05 μ s	4.64x
PyMatching v2.4 (C++)	PyMatching	25	1,000	6,730.2 dec/s	148.58 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	25	5,000	45,270.8 dec/s	22.09 μ s	1.23x
QECTOR Blossom (CPU)	QECTOR CPU	25	5,000	23,648.8 dec/s	42.29 μ s	0.64x
QECTOR CUDA Batch (GPU)	QECTOR GPU	25	5,000	165,830.9 dec/s	6.03 μ s	4.49x
PyMatching v2.4 (C++)	PyMatching	25	5,000	36,907.2 dec/s	27.09 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	25	10,000	79,828.7 dec/s	12.53 μ s	2.68x
QECTOR Blossom (CPU)	QECTOR CPU	25	10,000	24,143.6 dec/s	41.42 μ s	0.81x
QECTOR CUDA Batch (GPU)	QECTOR GPU	25	10,000	176,222.9 dec/s	5.67 μ s	5.91x
PyMatching v2.4 (C++)	PyMatching	25	10,000	29,806.3 dec/s	33.55 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	25	50,000	185,145.2 dec/s	5.40 μ s	3.32x
QECTOR Blossom (CPU)	QECTOR CPU	25	50,000	25,988.5 dec/s	38.48 μ s	0.47x
QECTOR CUDA Batch (GPU)	QECTOR GPU	25	50,000	93,461.1 dec/s	10.70 μ s	1.67x
PyMatching v2.4 (C++)	PyMatching	25	50,000	55,850.3 dec/s	17.90 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	25	100,000	197,734.8 dec/s	5.06 μ s	6.67x
QECTOR Blossom (CPU)	QECTOR CPU	25	100,000	23,603.7 dec/s	42.37 μ s	0.8x
QECTOR CUDA Batch (GPU)	QECTOR GPU	25	100,000	24,321.4 dec/s	41.12 μ s	0.82x
PyMatching v2.4 (C++)	PyMatching	25	100,000	29,651.6 dec/s	33.73 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	31	1,000	8,583.3 dec/s	116.51 μ s	3.31x
QECTOR Blossom (CPU)	QECTOR CPU	31	1,000	3,086.6 dec/s	323.98 μ s	1.19x
QECTOR CUDA Batch (GPU)	QECTOR GPU	31	1,000	23,648.2 dec/s	42.29 μ s	9.11x
PyMatching v2.4 (C++)	PyMatching	31	1,000	2,596.8 dec/s	385.09 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	31	5,000	2,082.8 dec/s	480.13 μ s	0.07x
QECTOR Blossom (CPU)	QECTOR CPU	31	5,000	4,198.6 dec/s	238.18 μ s	0.14x
QECTOR CUDA Batch (GPU)	QECTOR GPU	31	5,000	87,219.4 dec/s	11.47 μ s	2.84x
PyMatching v2.4 (C++)	PyMatching	31	5,000	30,717.2 dec/s	32.56 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	31	10,000	8,551.7 dec/s	116.94 μ s	0.19x
QECTOR Blossom (CPU)	QECTOR CPU	31	10,000	4,596.3 dec/s	217.57 μ s	0.1x
QECTOR CUDA Batch (GPU)	QECTOR GPU	31	10,000	34,057.9 dec/s	29.36 μ s	0.77x
PyMatching v2.4 (C++)	PyMatching	31	10,000	44,072.3 dec/s	22.69 μ s	1.0x

Decoder	Category	d	Shots	Throughput	Latency	Speedup
QECTOR Fast Union-Find (CPU)	QECTOR CPU	31	50,000	24,144.3 dec/s	41.42 μ s	0.56x
QECTOR Blossom (CPU)	QECTOR CPU	31	50,000	5,268.6 dec/s	189.80 μ s	0.12x
QECTOR CUDA Batch (GPU)	QECTOR GPU	31	50,000	21,178.1 dec/s	47.22 μ s	0.5x
PyMatching v2.4 (C++)	PyMatching	31	50,000	42,771.6 dec/s	23.38 μ s	1.0x
QECTOR Fast Union-Find (CPU)	QECTOR CPU	31	100,000	86,928.9 dec/s	11.50 μ s	2.15x
QECTOR Blossom (CPU)	QECTOR CPU	31	100,000	6,780.2 dec/s	147.49 μ s	0.17x
QECTOR CUDA Batch (GPU)	QECTOR GPU	31	100,000	21,502.9 dec/s	46.51 μ s	0.53x
PyMatching v2.4 (C++)	PyMatching	31	100,000	40,436.7 dec/s	24.73 μ s	1.0x